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Power Stabilization for Large-Scale AI Workloads

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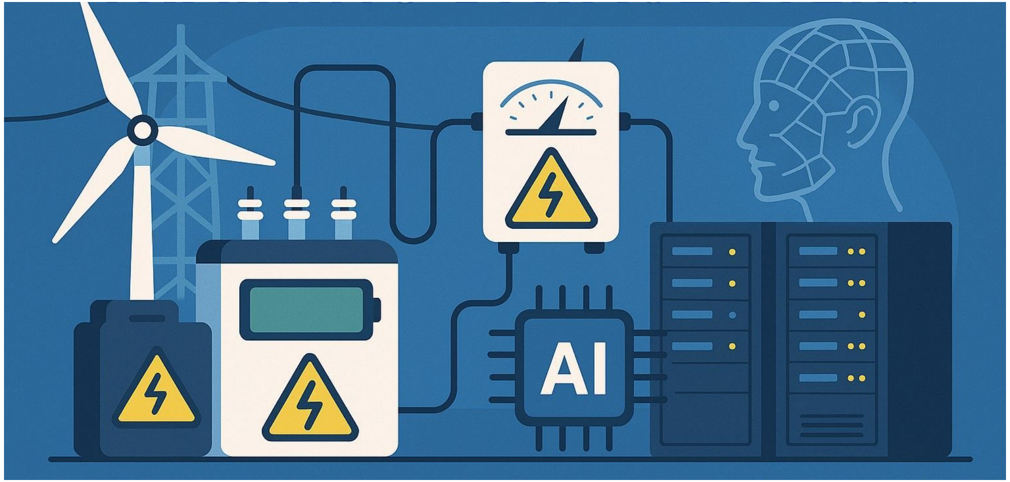
Power Stabilization for AI Training Datacenters

Microsoft, OpenAI and NVIDIA published on arXiv

Previously on CSE 585…

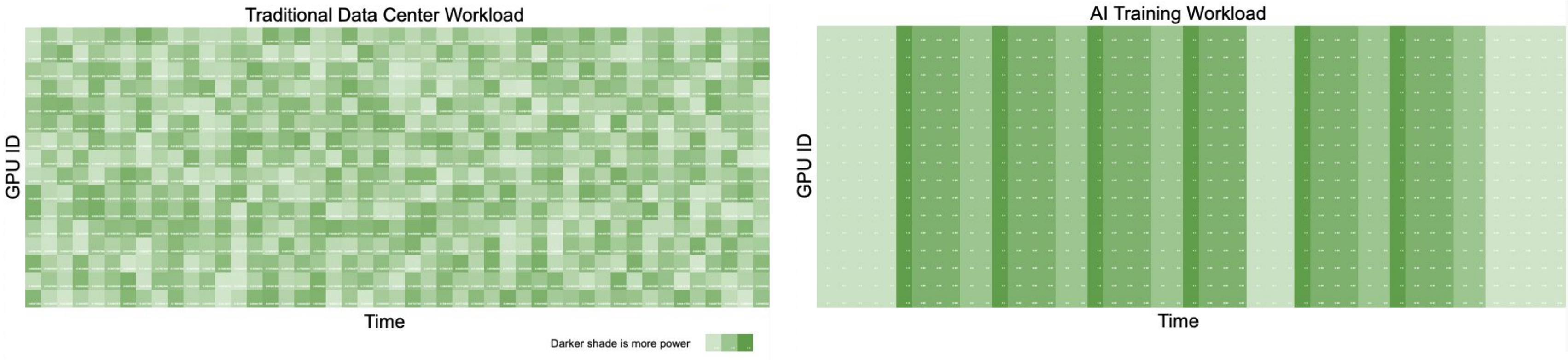
- We saw systems optimized for energy consumption and efficiency:
    - Perseus
    - Kareus
    - TAPAS

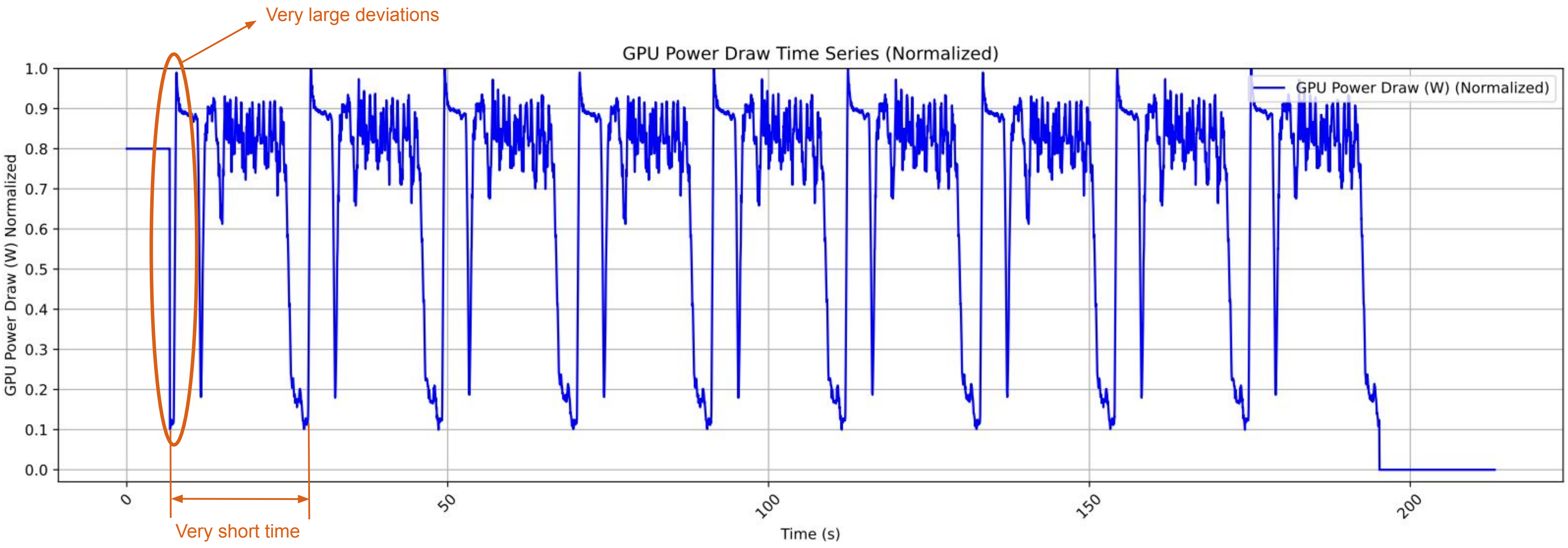


- Today's agenda: **POWER STABILIZATION**

Problem at Hand

- Single training job uses >100,000 GPUs
- Compute-heavy and communication-heavy phases utilize very different amounts of power
- Causes huge power swings at node-level
- Visible at rack, datacenter and power-grid levels due to synchronous nature
- Power consumption can even oscillate by tens of megawatts within a single datacenter
- Resonance can cause mechanical failure




Very large deviations
GPU Power Draw Time Series (Normalized)
GPU Power Draw (W) (Normalized)
GPU Power Draw (W) Normalized
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
50
100
150
200
Time (s)
Very short time

Why is this a problem?

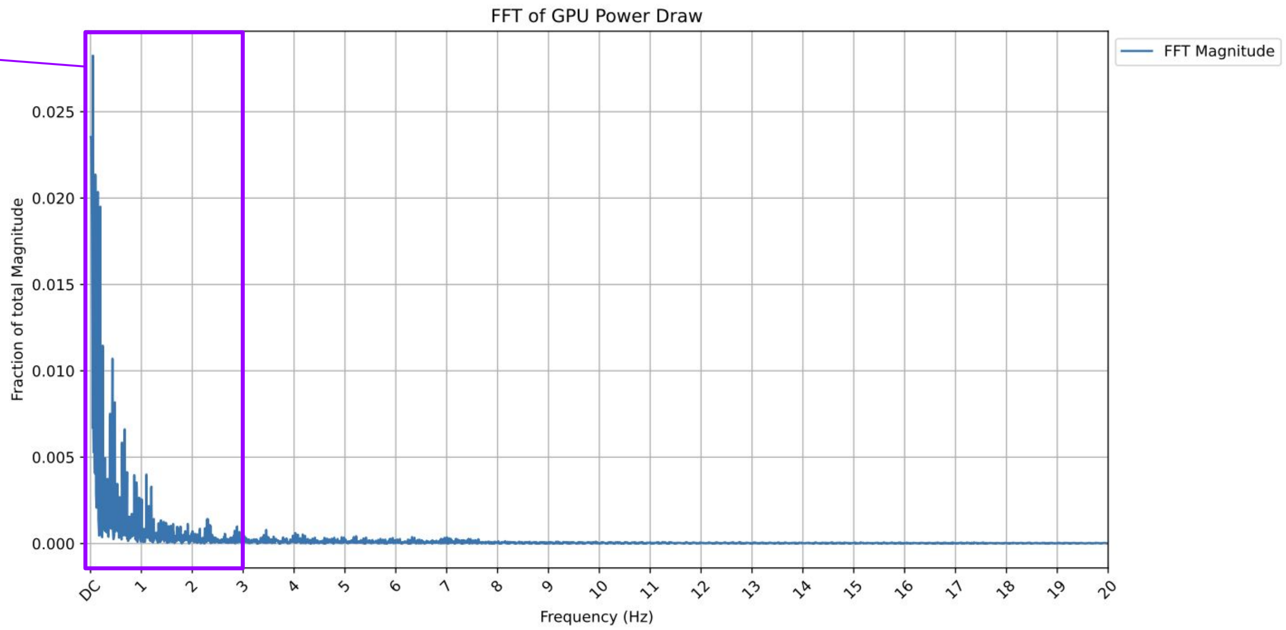
- Power Distribution Units undergo mechanical stress
  - High current leads to magnetic forces which cause fatigue and loose connections
- Swings in power lead to magnetic core reaching saturation → inefficient and less durable
- Voltage fluctuations impact grid frequency regulation in isolated regions

More problems

- **Mechanical Resonance:** Causes turbine shafts to twist (torsion) at high magnitudes at resonant frequencies
- **Sub-Synchronous Resonance:**
  - Power fluctuation triggers the transmission line's resonant frequencies
  - Results in a feedback loop that amplifies torsional oscillations resulting in shaft fatigue

Consequences

Larger fractions of the total magnitude have frequencies in the critical range



What does this paper propose?

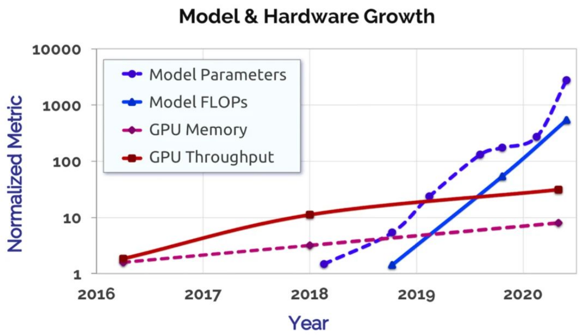
Ways to mitigate risks at 3 levels:

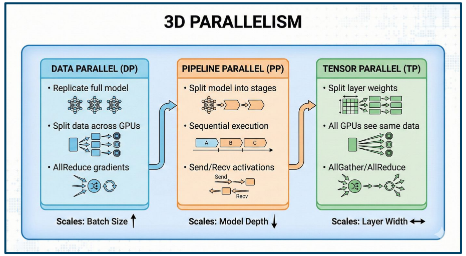
- **Software:** Injecting controlled workloads to smooth out transitions
- **GPU-level:** Enforcing ramping constraints and power floors
- **Infrastructure:** Energy storage at rack-level, to absorb and release power as needed

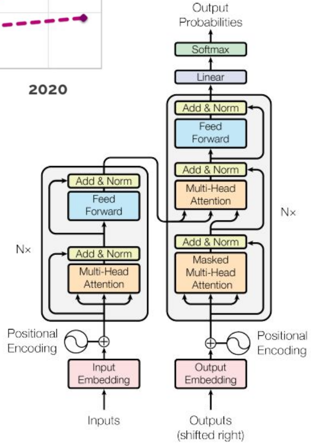
Call-to-Action: Cross industry co-design across software, hardware and infrastructure so that AI systems are scalable and more power-aware

Rise of Large Training Jobs

- **Scaling of AI models driven by trends like:**
  - Transformers
  - 3D parallelism (data, tensor, pipeline)
  - Hardware evolution (higher bandwidth interconnects – NVLink, InfiniBand)
  - Larger datacenter deployment size
- Models with parameters of the order of billions demand large computational resources
- **Ways to enable cheaper training runs:**
  - Domain targeted training
  - Model size efficiency
  - Pretrained backbones
- Foundation model training spans 10,000s of GPUs

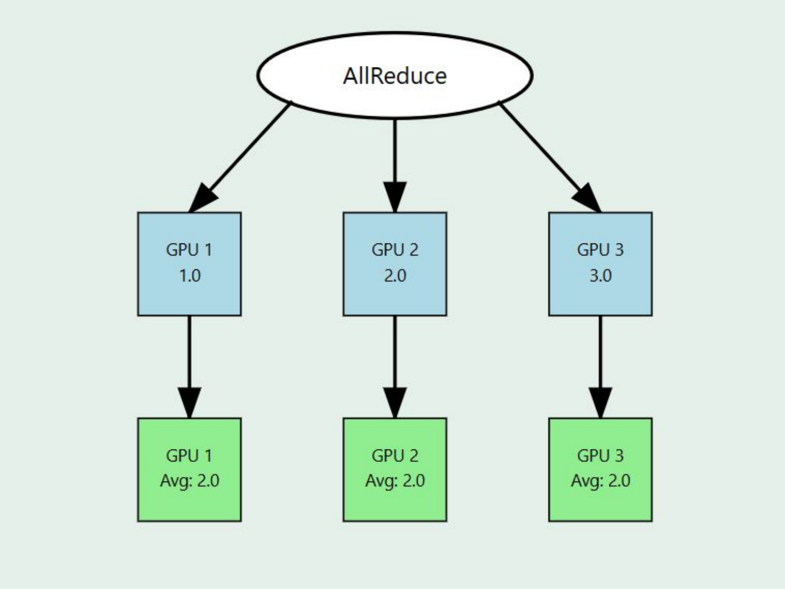






Compute and Communication Phases

- AI training has 2 distinct phases
  - Compute
    - Forward pass
    - Backward pass
  - Communication
    - All-reduce
    - Checkpointing (model parameters, optimizer states)

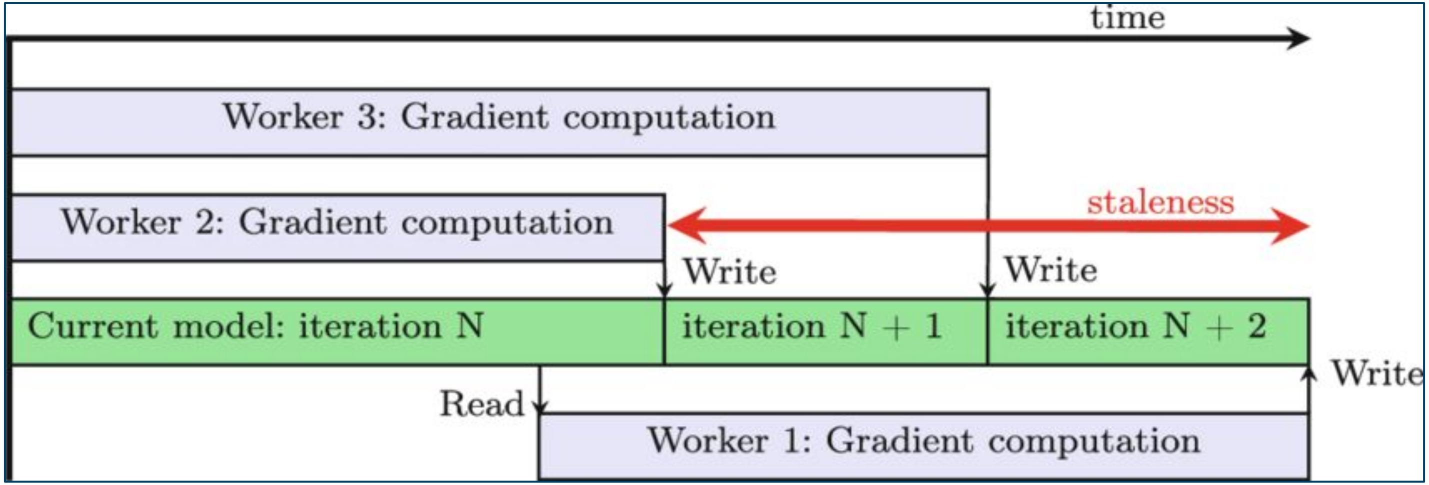


Potential Solution with Tradeoffs

Asynchronous training methods with lazy weight updates

- **Accuracy tradeoff:** Stale gradients
- **Convergence tradeoff:** Convergence rate $O(1/\sqrt{K})$ → inefficient

Total number of update steps or iterations taken by the model during training



Some Terminology to Know

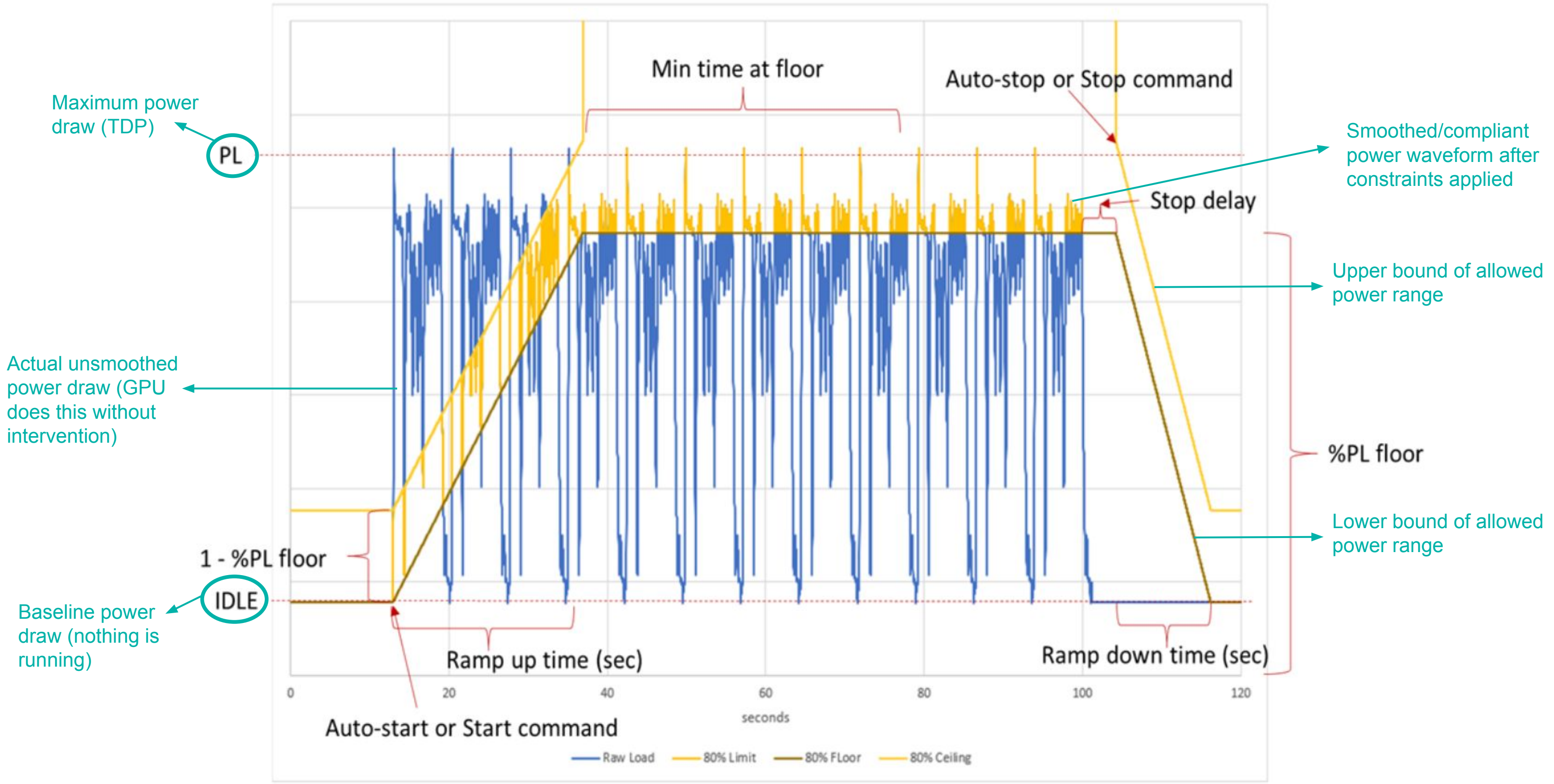
- **Thermal Design Power (TDP)**
  - Maximum amount of heat a component (like a GPU) is designed to dissipate under a normal, heavy workload
  - In training, compute-heavy phases typically run at or near this limit
- **EDP (Electrical Design Power)**
  - Short-term power overshoots (lasting ~50ms) that occur when a workload spikes up
- **MPF (Minimum Power Floor)**
  - Even if the GPU is idle or waiting for data, it consumes this baseline amount of energy to keep the load flat
  - Current limits usually allow an MPF of up to 90% of TDP

Some Terminology to Know

- **PDU (Power Distribution Unit)**
  - High-tech power strip for the datacenter
  - Takes the massive amount of electricity coming into the building and distributes it to individual racks and servers
- **UPS (Uninterruptible Power Supply)**
  - It provides near-instantaneous protection from power interruptions and cleans up voltage spikes or sags

Utility-Level Specifications

1. <u>Time-domain spec</u>
   a. **Ramp-up rate (MW/s)** → maximum permitted rate of increase in power usage
   b. **Ramp-down rate** → maximum permitted decrease in power consumption over time
   c. **Dynamic power range** → Allowed short-term deviation in power consumption before ramp constraints are triggered


Maximum power draw (TDP)
PL
Actual unsmoothed power draw (GPU does this without intervention)
1 - %PL floor
IDLE
Baseline power draw (nothing is running)
Min time at floor
Auto-stop or Stop command
Smoothed/compliant power waveform after constraints applied
Stop delay
Upper bound of allowed power range
%PL floor
Lower bound of allowed power range
Ramp up time (sec)
Ramp down time (sec)
Auto-start or Start command
0
20
40
60
80
100
120
seconds
Raw Load
80% Limit
80% FLoor
80% Ceiling

Utility-Level Specifications

2. <u>Frequency-domain spec</u>
    a. **Critical frequency range** → Eg. 0.1–20Hz
    b. **Maximum allowed spectral magnitude** → if frequency within critical frequency range, amplitude should be reduced (20% of total harmonic energy within that range)

Understanding the spec..

- **<1Hz**
  - Long transmission lines connecting independently strong portions of the grid
  - Damping ratios are greater than 1 (desirable)
  - Danger of instability during periodic load variations
- **1Hz to 2.5Hz** → Closely coupled sources (units in the same plant or plant-to-plant in close proximity)
- **7Hz to >100Hz** → Shaft torsional critical frequencies

Additional Requirements

1. **Ability to meet different utility specifications**
2. **Minimal performance loss** of load-shaping or smoothing mechanisms
3. **Minimal wasted energy**
4. **Control over EDP** so that peaks aren't visible beyond rack supply units

Mitigation Strategies

An ideal strategy to deal with power swings:

- ➢ Can meet the tightest of time and frequency domain specs
- ➢ Is reliable and durable
- ➢ Has good performance for low energy and cost

Mitigation Strategies

Authors look at 3 types of solutions:

- ➢ Software-only
- ➢ GPU Power Smoothing
- ➢ Rack-Level Energy Storage

Software-Only Mitigation

- Dynamically introduce power-hungry secondary workloads when GPU activity falls below threshold

- System obtains uniform power draw mitigating the swings

Concerns to Address

Type of Secondary Workload: 2 choices

1. Low-priority, useful job like data augmentation for preprocessing
    a. Need to save job state
    b. Larger context switching overhead
    c. Might impact primary workload performance

2. Artificial (Dummy) workload like General MatMul
    a. Wastes energy
    b. Might impact primary workload performance

Concerns to Address Contd

Monitoring:

- ➢ Secondary workload cannot be deterministically called
- ➢ Need a software mechanism that uses real-time, fine-grained power and activity monitoring of the GPUs
- ➢ Even reliable 100ms activity counters are too slow

Concerns to Address Contd

Secondary Workload Start/Stop

- ➢ Start as soon as block activity drops below threshold
- ➢ When primary ramps up again, secondary needs to back-off
- ➢ No activity counters per process → secondary workload periodically backs-off

But why block activity?

- NVIDIA GPUs break data and performs compute across thread blocks
- The scheduler allocates the blocks to Streaming Multiprocessors
- Monitoring block activity measures how many blocks are active and more active blocks in SMs -> compute phase

Software-Only Solution: FIREFLY

- Use NVIDIA's Multi-Process Service (MPS) which allows multiple CUDA processes to share a single GPU context

- Makes context switching faster

- Helps to minimize the impact on the primary workload

But why NVIDIA MPS?

- MPS is a runtime feature that allows multiple CPU processes to share a single GPU's resources
- It runs tasks concurrently and shares the same GPU context
- Reduces time needed for context switching by using spatial switching instead of time slicing

Firefly Design Choices

- Uses MatMul for secondary workload
- Block Activity counters for the GPU for monitoring
- Successfully increased power utilization to 100% of TDP

Challenges

- **Performance Overheads:** Secondary workload needs compute and memory; backoff mechanism still impacts primary workload performance
- **Reliability:** Failure domain of primary and secondary are tied resulting in fault propagation
- **Energy Wastage**

But what are failure domains?

- ➢ Its a specific section of a computing network, infrastructure, or physical site that is negatively affected when a critical component fails (Blast radius)
- ➢ In our context, if secondary triggers a GPU error, the GPU resets and since the context is shared with the primary, it is also affected

Potential Improvements

- Separate Failure Domains: Something other than MPS? Proactive fault detection strategies?
- Replace monitoring with predictive modelling of compute/communication phases
- Hardware vendor can provide tightly coupled software solutions with the scheduler and monitoring
- Priority scheduling instead of GPU preempting the secondary

Next Solution: GPU Power Smoothing

- Use the GPU Power Smoothing feature from NVIDIA GB200 superchip
- **GB200 -** Flattens power spikes using
  - Electrolytic power capacitors to charge in valleys and discharge in peaks
  - Do a GPU burn by engaging hardware power burners
  - Has programmable ramp rates and Minimum Power Floor to manage GPU power consumption

GB200 GPU Profile

The programmable GPU profile has

- **Ramp-up and ramp-down rates** directly from time domain specs
- **Minimum Power Floor** sets floor of power consumption and allows us to meet DPR specs
- **Stop Delay** specifies how long GPU should be in PF before ramping down (trade-off between performance impace and energy burn)

Some Results

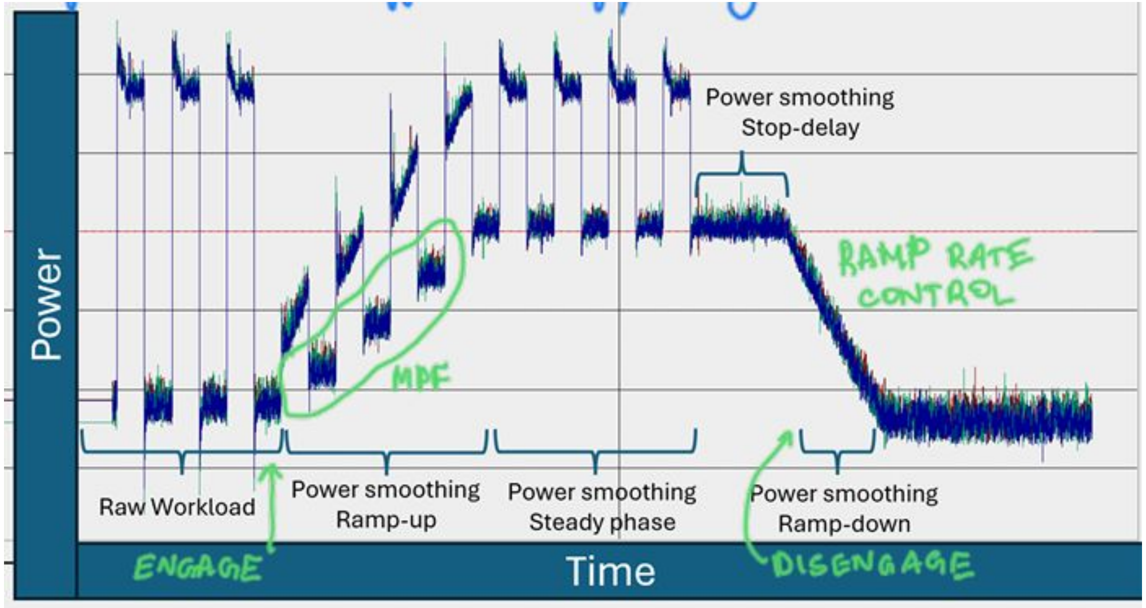


Fig. GB200 Power smoothing results with a square-wave microbench mark.

- ➢ MPF setting prevents a massive valley
- ➢ Ramp Rate Control ensures ramp down isn't a vertical drop and is smooth
- ➢ Workload stays the same throughout, until the stop delay phase, where the workload has no activity

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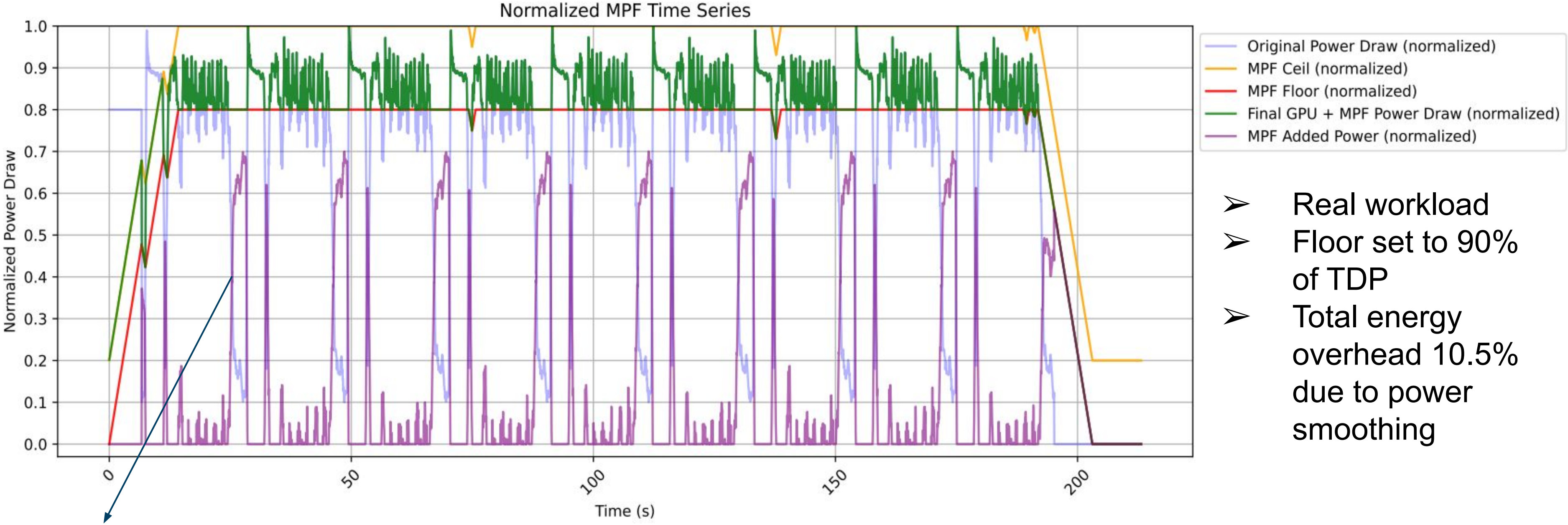

- Real workload
- Floor set to 90% of TDP
- Total energy overhead 10.5% due to power smoothing

Power added by GB200

Fig . Power smoothing to the MPF simulated on the training waveform from Fig 1

Challenges

- Additional energy burnt
- GB200 has a 5 year lifespan but not sure if it will endure that long
- Cannot handle strict DPR specs: Can support only MPF of 0.9x TDP and minimum EDP of 1.1x TDP meaning minimum DPR is 20%

Next Solution - Energy Storage

Best case is an Energy Storage Solution at **rack-level** that

- ➢ Can directly measure the load
- ➢ Has enough capacitance to support the workload
- ➢ Can meet the power fluctuations
- ➢ Can quickly switch between charging and discharging

Solution charges during communication phase and discharges during compute phase

But why rack-level?

- Options are server-level, rack-level, row-level, datacenter-level
- Placing at higher level exposes more devices like PDUs and UPSes to perturbations without any notable gains
- Placing at a higher level makes the battery a single point of failure
- AC-DC converters are in the rack-level
- **Therefore, rack-level is the best option**

Simulated Working of this Solution

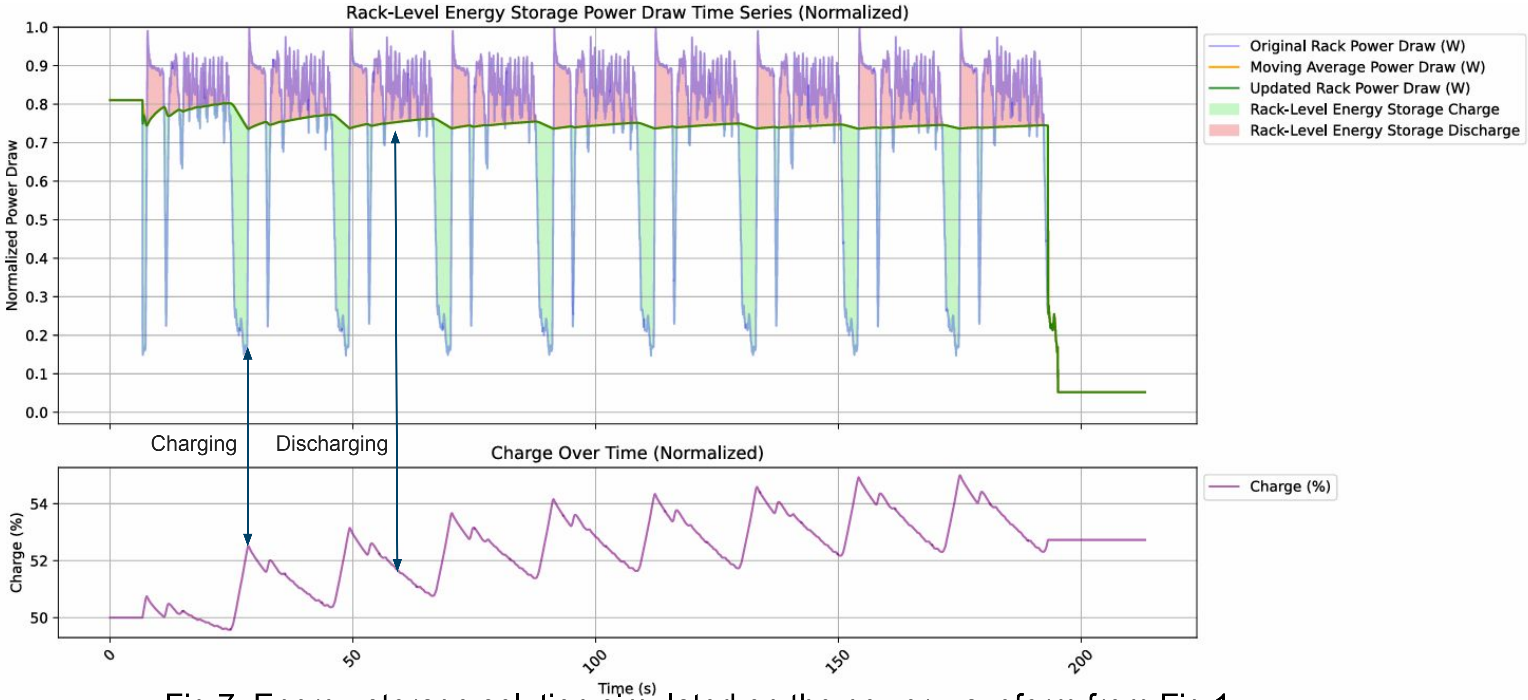

Fig 7. Energy-storage solution simulated on the power waveform from Fig 1

Challenges

- ➢ Frequency of power swings form a large spectrum
- ➢ A single energy solution is difficult - easier for higher frequencies but lower frequencies are tricky
- ➢ Ramping requires very large capacitance but occurs rarely - doesn't necessarily pay off

Summary of Solutions

Solution	Reliability	Performance	Energy	Cost	Ability to Meet Tightest Spec	Dependency on the Developer	Lifetime
Software-only mitigation	Medium	Medium	High	Medium	High	High	High
GPU power smoothing	High	Medium	High	Low	Medium	Medium	Medium
Rack-level energy storage	High	High	Low	High	High	Low	High

Building an Ideal Solution

- Combination of solutions -GPU level power smoothing using software/hardware solutions
- Rack-level energy storage
- Real time telemetry monitoring to detect resonance
- BESS (Battery Energy Storage Systems) at larger scales
- Solution requires communication between different solutions regarding state of charge

Fast Telemetry-Based Backstop

- This system monitors power waveforms across the datacenter
- Looks for early signs of resonance using **lower-latency telemetry** and **real-time FFT-bin monitoring**
- Intervenes by **soft throttling** or **load shaping**
- If ineffective, use more aggressive responses like **circuit-level power shedding** or **coordinated disconnects**

But what do these words mean?

- Low latency telemetry collects data at 1ms interval
- Real time FFT-bin monitoring monitors the frequency of these power swings in real time
- Soft throttling adaptively reduces the GPU power usage (like the predictive modelling solution)
- Load Shaping is power smoothing
- Circuit level power shedding basically cuts off the power to a rack in the datacenter if they see its load spiking too dangerously
- Coordinated disconnects are strategic scale downs on the amount of GPUs working on an AI workload

Strengths

- **Multi-Pronged Mitigation**: Software, Firmware, and BESS consideration that provides a versatile toolkit for different utility requirements.
- **Open Standards Advocacy:** Call to Action via Open Compute Project advocates for interoperable power telemetry.
- Incentivizes AI and system engineers to build **algorithms moving away from bulk synchronous parallelism**

Limitations

- ➢ **Energy Inefficiency**: 10.5% energy overhead at 90% MPF. Increases carbon footprint and electricity costs significantly. **Is saving power swings at the expense of more energy worth it?**
- ➢ **Hardware Constraints**: 20% residual dynamic range in GB200. Cannot meet the tightest utility specs without secondary systems.
- ➢ **Infrastructure CapEx**: High physical/cost footprint of BESS. Space and safety constraints limit brownfield deployment.

Call to Action

To handle power stabilization, we need

1. AI Frameworks and System Designers: Design **async training techniques**, staggered scheduling, and overlap of compute and communication
2. Utility and Grid Operators: Openly s**hare resonance and ramp specs**
3. **Industry Collaboration to establish interoperable standards** for telemetry, load signaling, and sub-synchronous oscillation mitigation.

Industry’s Response (Future Directions)

- Asynchronous & Power-Flat Training Algorithms
- Hybrid Energy Storage Systems (HESS)
- Ultra-Low Latency Telemetry & Backstop Systems

Industry's Response (Future Directions)

- Asynchronous & Power-Flat Training Algorithms
  a. Move beyond BSP → eliminate synchronicity at the root
  b. Make AD-PSGD and lazy weight update methods robust at 100k-GPU scale
  c. Goal: naturally smooth power profile without energy waste like Firefly or MPF
- Hybrid Energy Storage Systems (HESS)
  a. Pair high-power devices (supercapacitors, flywheels) with high-energy batteries (LFP, LTO)
  b. No single technology meets combined needs of high ramp rates, frequent cycling, and long discharge
  c. Develop AI-aware degradation models → AI training's jitter-like spikes shorten battery cycle life

Industry's Response (Future Directions)

- Ultra-Low Latency Telemetry & Backstop Systems
  a. Current reliable counters at 100ms → too slow for 20Hz+ oscillation detection
  b. Need sub-millisecond telemetry with real-time spectral analysis
  c. Integrate with circuit-level power shedding faster than any human operator or grid breaker
- Global Interoperable Datacenter-to-Grid Standards
  a. Open-source load signaling frameworks → datacenters communicate ramp schedules to utilities in real time
  b. Integrate with existing Energy Management Systems used by grid operators
  c. Critical for ensuring AI scaling does not compromise civilian power infrastructure

Kareus, Perseus and TAPAS

- They improve energy efficiency whereas we are concerned about energy stability.
- Not always orthogonal to optimize — Perseus might create a power "dip" to save a few cents of electricity, which Firefly would immediately detect as a "valley" and fill with dummy work
- **Perseus and Kareus** lower the amplitude of the swings making it easier to match the spectral magnitude specs but frequency alteration can cause smaller high frequency power fluctuations
- **TAPAS** changes the frequency of the swings and will alter the nature of the pulses

Conclusion

- The synchronous and periodic nature of AI training workloads cause power swings
- If the frequency of these power swings match resonant frequencies of utilities, the utilities can go down
- The paper explores mitigation strategies and identifies challenges
- Calls for industry wide collaboration on interoperable standards, low-latency telemetry and different AI training algorithms

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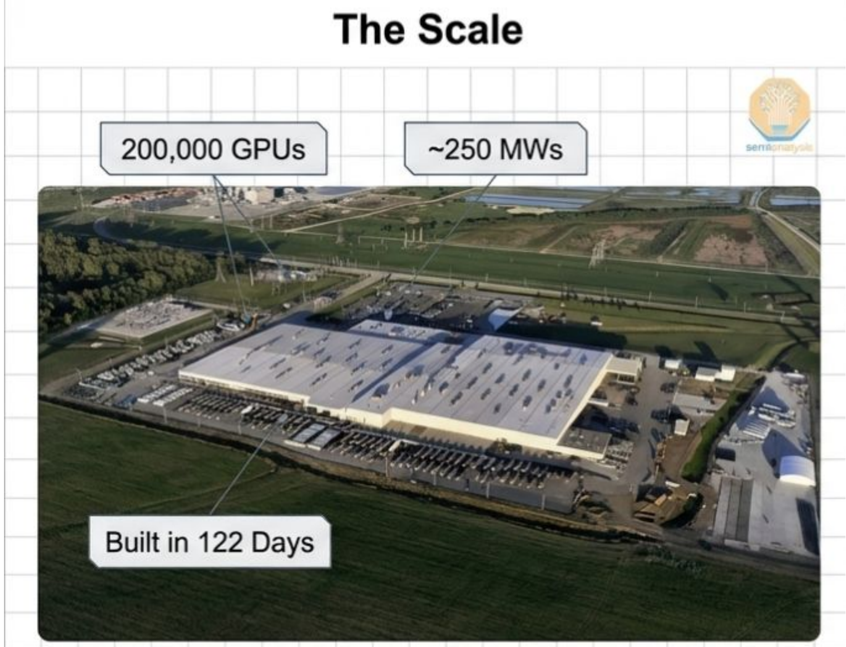
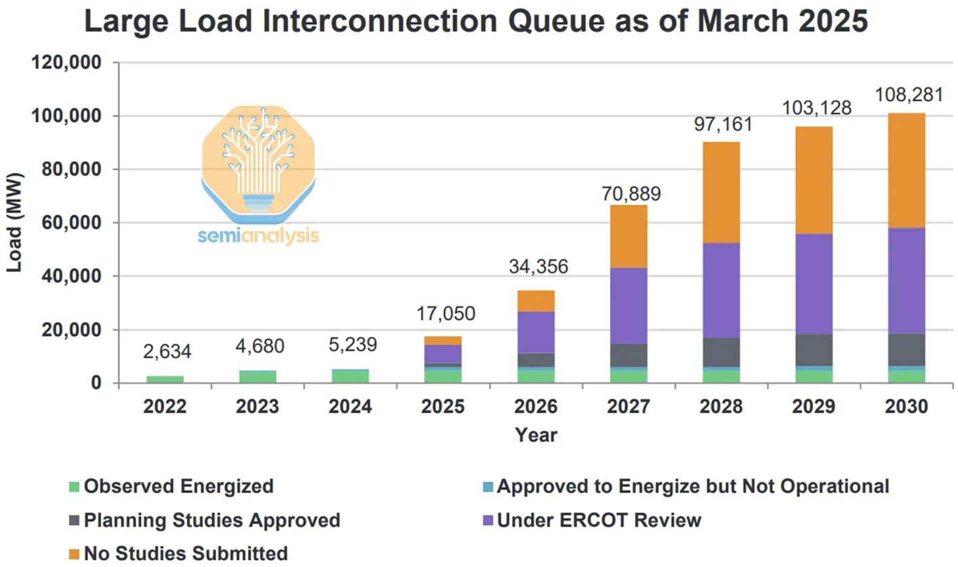
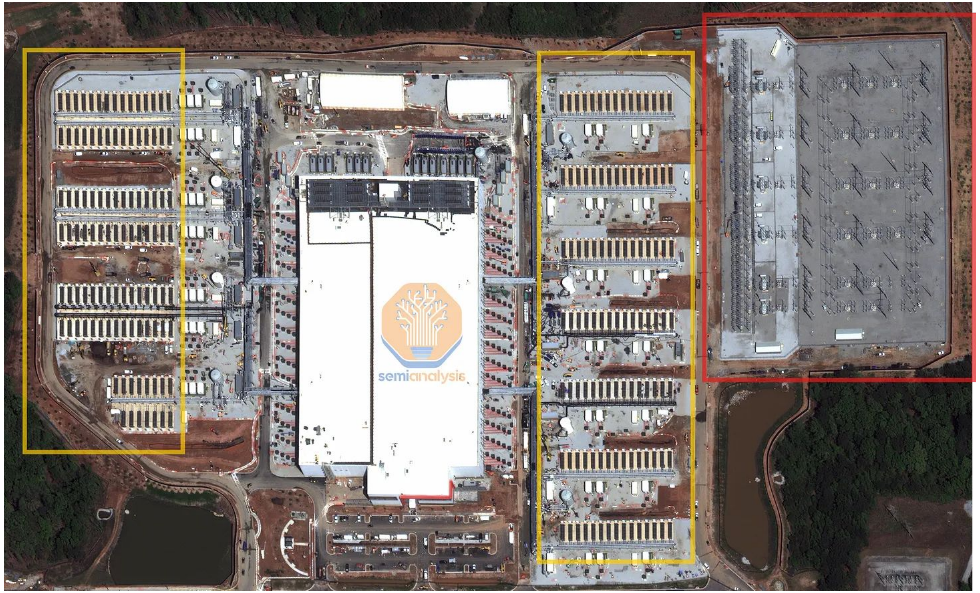
AI Training Load Fluctuations at Gigawatt-Scale – Risk of Power Grid Blackout?

The Giga watt Scale Reality





The Scale of the Problem



- OpenAI's key training cluster: ~300 MW IT capacity, ~400 MW nameplate — world's largest single building
- Second identical building under construction since Jan 2025 → campus reaches gigawatt-scale by mid-2026
- ERCOT large load queue: 108 GW (majority: datacenters)
- US peak load for context: 745 GW
- NERC asking all major transmission utilities how they model datacenter loads

The Problem in One Sentence

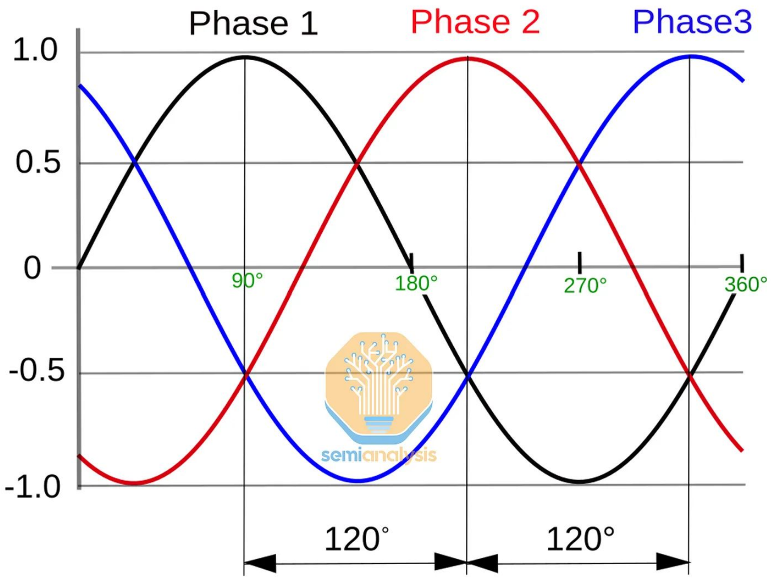
> “AI training workloads unexpectedly rise and fall from full load to nearly idle in fractions of a second. At gigawatt-scale, the worst-case scenario is a blackout for millions of Americans.”
>
> — SemiAnalysis, June 2025

108 GW in ERCOT's large load queue

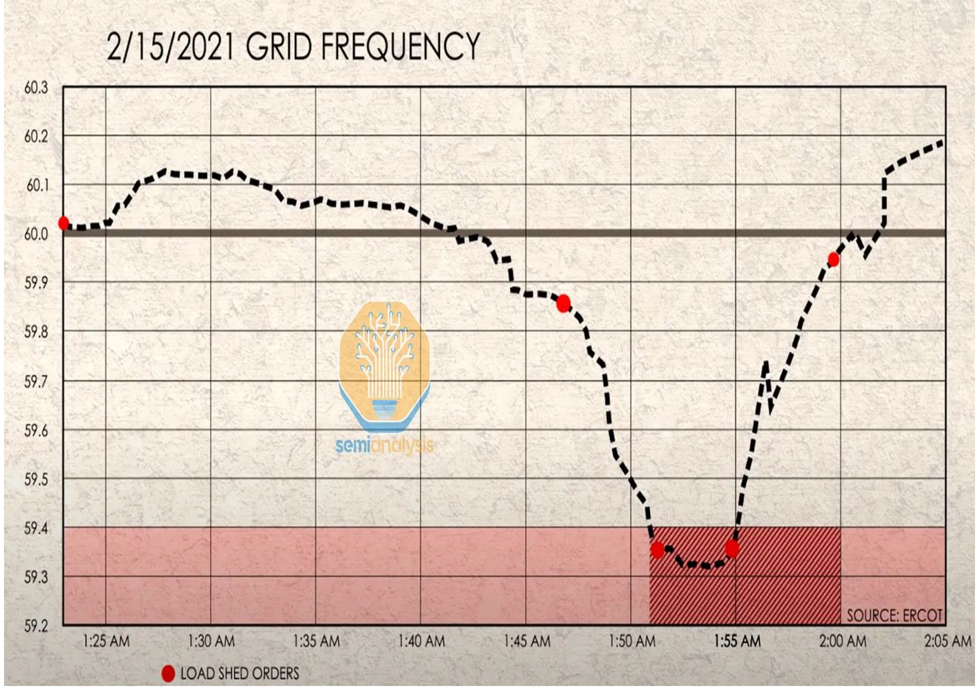
~30 MW: Meta LLaMa 3 cluster (24,000 H100s)

pytorch_no_powerplant_blowup=1
tens of millions/year wasted

Power quality

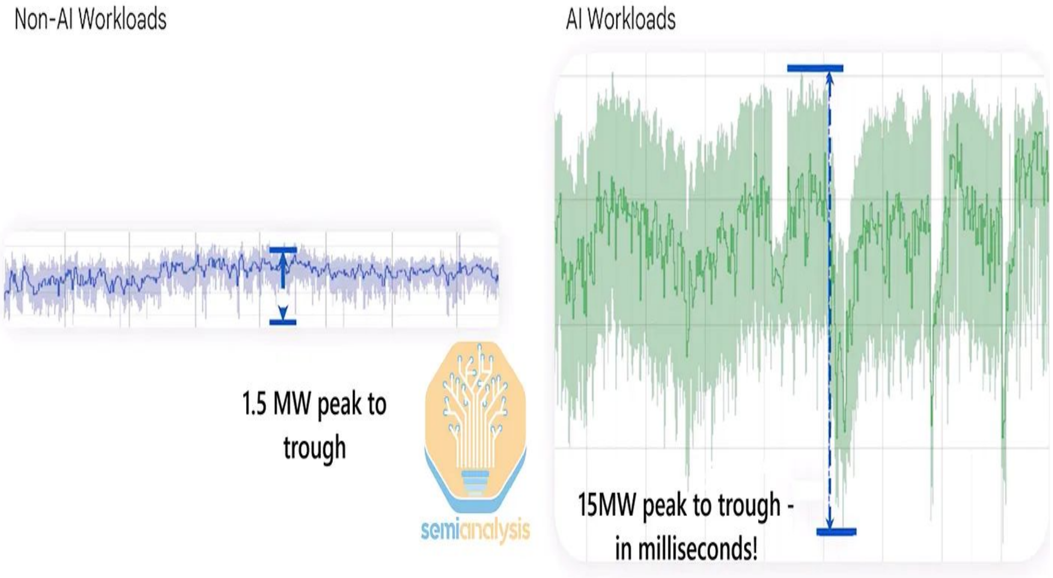


- AC grid: 60 Hz (North America), 50 Hz (Europe/Asia)
- 3 wires, each offset by 120° — that's three-phase power
- Datacenters run on three-phase, not standard residential single-phase
- All three phases must stay balanced — imbalance = instability
- Supply/demand mismatch → waveform drifts → motors fry, breakers trip
- Spinning generators historically kept this stable via rotational inertia
- Renewables + AI datacenters break this chain — no rotating mass, no passive stabilization



- Texas 2021 : Frequency fell below 59.4 Hz → 9 min at that level = multi-day blackout
- ERCOT response: forced load shed on homes and businesses
- A GW-scale AI datacenter tripping offline can cause the same excursion
- Single load event, not a generation failure

Cloud Datacenters Vs AI Clusters



- Cloud VMs are heterogeneous → power naturally averages out
- AI GPUs are synchronous → spikes and drops are fully correlated
- 15× gap understates the real risk at gigawatt scale
- Grid operators' tools were never designed for this load profile

- 200k GPUs, ~250 MW, built in 122 days in Memphis
- A single checkpoint sync = tens of MW instantaneous drop
- Now the reference architecture for the industry

how may homes can run in 250 MW of power in Mephisis?

1. **Average Memphis Household Usage:** A typical Memphis home uses about **1,467 kWh per month** (or ~17,600 kWh per year).
2. **Continuous Power of 1 Home:** To find out how many "Watts" a home uses on average at any given moment, we divide the monthly usage by the number of hours in a month:

$$\frac{1,467\ \text{kWh}}{730\ \text{hours}} \approx 2.0\ \text{kW per home}$$

3. **Total Capacity:** We take the 250 MW (250,000 kW) and divide it by the average draw per home:

$$\frac{250,000\ \text{kW}}{2.0\ \text{kW/home}} = 125,000\ \text{to}\ 155,000\ \text{homes}$$

what if the plant consumes 250 MW running 24*7 for a month in Mephisis . how much is my bill

1. Energy Consumption (The "Usage")

First, we calculate how many Megawatt-hours (MWh) you consume in a typical 30-day month:

$$250\ \text{MW} \times 24\ \text{hours} \times 30\ \text{days} = 180,000\ \text{MWh}$$

(This is equivalent to 180 million kWh).

2. The Bill Breakdown (Estimated)

Your bill consists of two primary parts: **Demand Charges** (paying for the capacity) and **Energy Charges** (paying for the actual fuel/electrons).

Component	Rate Estimate (2026)	Monthly Cost
Demand Charge	~$18.00 per kW	**$4,500,000**
Energy Charge	~$0.054 per kWh	**$9,720,000**
Service Fees/Fixed	Negotiated	~$1,000+
Total Estimated Monthly Bill		**~$14,220,000**

The AI training -in action

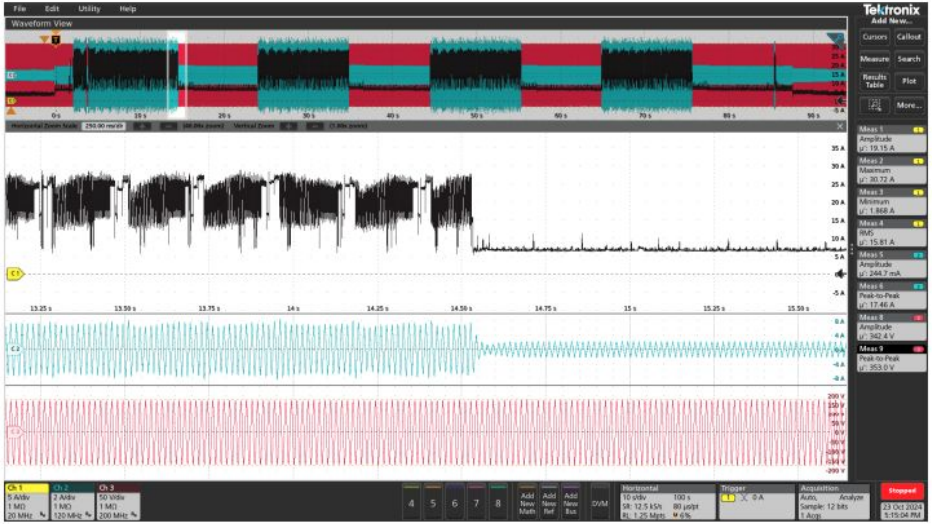
C:\WINDOWS\system32\cmd.

```
Tue Apr  7 01:58:02 2026
+-----------------------------------------------------------------------------------------+
| NVIDIA-SMI 595.79                 Driver Version: 595.79         CUDA Version: 13.2     |
+-----------------------------------------+------------------------+----------------------+
| GPU  Name                  Driver-Model | Bus-Id          Disp.A | Volatile Uncorr. ECC |
| Fan  Temp   Perf          Pwr:Usage/Cap |           Memory-Usage | GPU-Util  Compute M. |
|                                         |                        |               MIG M. |
|=========================================+========================+======================|
|   0  NVIDIA GeForce RTX 3060 ...   WDDM |   00000000:01:00.0  On |                  N/A |
| N/A   86C    P0            131W /  138W |    4686MiB /   6144MiB |     97%      Default |
|                                         |                        |                  N/A |
+-----------------------------------------+------------------------+----------------------+

+-----------------------------------------------------------------------------------------+
| Processes:                                                                              |
|  GPU   GI   CI              PID   Type   Process name                        GPU Memory |
|        ID   ID                                                               Usage      |
|=========================================================================================|
|    0   N/A  N/A            5864    C+G   C:\Windows\explorer.exe                  N/A      |
|    0   N/A  N/A           14116    C+G   ...1g1gvanyjgm\WhatsApp.Root.exe         N/A      |
|    0   N/A  N/A           14612    C+G   ...ommon\Call of Duty HQ\cod.exe         N/A      |
|    0   N/A  N/A           22604    C+G   ...s\PowerToys.ColorPickerUI.exe         N/A      |
+-----------------------------------------------------------------------------------------+

Waiting for 1 seconds, press a key to continue ...
```

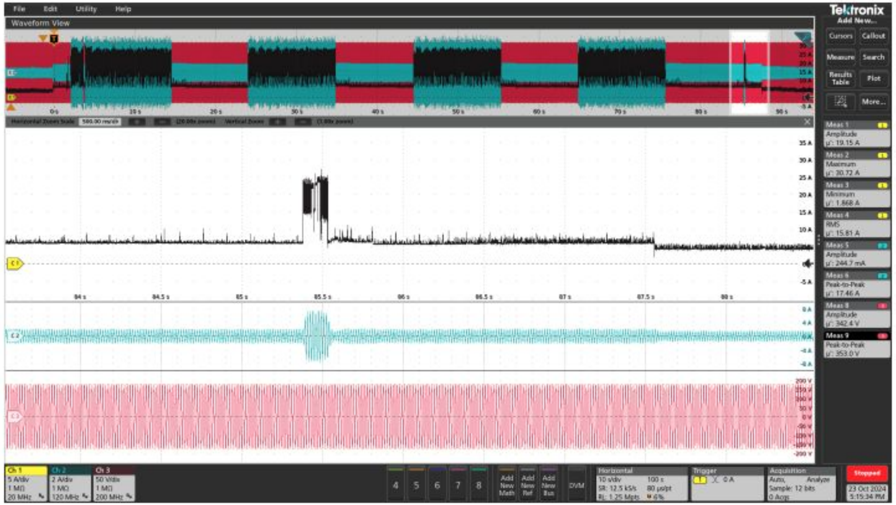
C:\WINDOWS\system32\cmd.

```
Tue Apr  7 01:59:09 2026
+-----------------------------------------------------------------------------------------+
| NVIDIA-SMI 595.79                 Driver Version: 595.79         CUDA Version: 13.2     |
+-----------------------------------------+------------------------+----------------------+
| GPU  Name                  Driver-Model | Bus-Id          Disp.A | Volatile Uncorr. ECC |
| Fan  Temp   Perf          Pwr:Usage/Cap |           Memory-Usage | GPU-Util  Compute M. |
|                                         |                        |               MIG M. |
|=========================================+========================+======================|
|   0  NVIDIA GeForce RTX 3060 ...   WDDM |   00000000:01:00.0  On |                  N/A |
| N/A   55C    P8             13W /  140W |     347MiB /   6144MiB |      1%      Default |
|                                         |                        |                  N/A |
+-----------------------------------------+------------------------+----------------------+

+-----------------------------------------------------------------------------------------+
| Processes:                                                                              |
|  GPU   GI   CI              PID   Type   Process name                        GPU Memory |
|        ID   ID                                                               Usage      |
|=========================================================================================|
|    0   N/A  N/A            5864    C+G   C:\Windows\explorer.exe                  N/A      |
|    0   N/A  N/A           14116    C+G   ...1g1gvanyjgm\WhatsApp.Root.exe         N/A      |
|    0   N/A  N/A           22604    C+G   ...s\PowerToys.ColorPickerUI.exe         N/A      |
+-----------------------------------------------------------------------------------------+

Waiting for 1 seconds, press a key to continue ...
```

The AI training- In clusters



(a) Initial capture showing the first load drop around 14.5 s. Checkpoints induce a sudden drop from peak current to near-idle, underscoring the abrupt nature of AI training workload transitions.



(b) Closer view (10 s scale) of the last checkpoint event, where training is intentionally interrupted, causing rapid up/down current surges.

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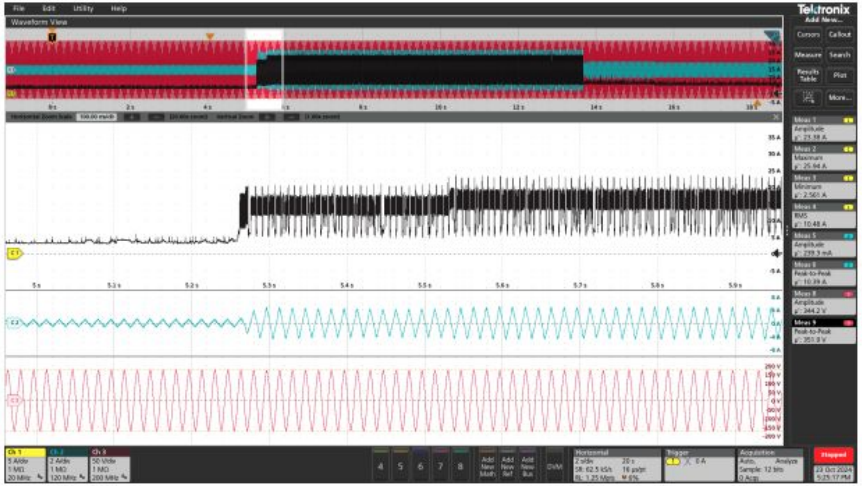
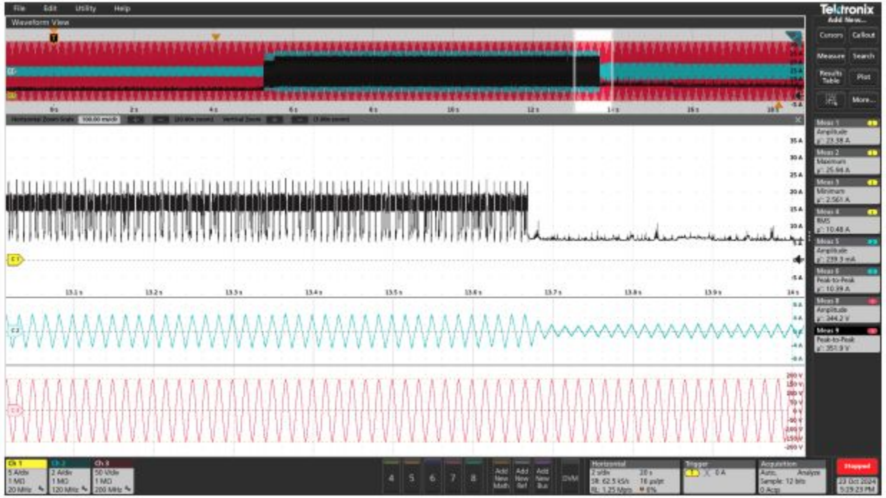
The AI Inferencing - In clusters



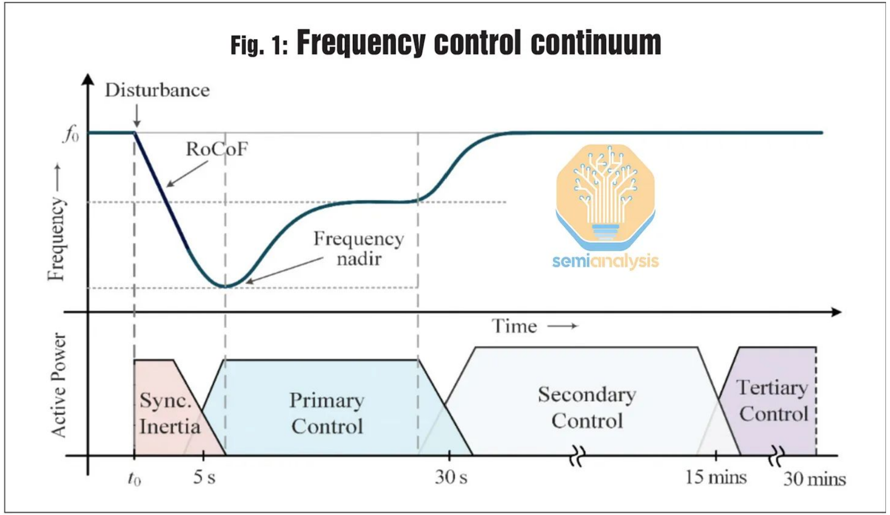
(a) Initial capture showing the inference load-up transition for an RTX-4090 running a LLaMA-3.1 8B model. The GPU current (black trace) rapidly increases, indicating the onset of inference operations.



(b) Subsequent capture illustrating the initial portion of the load-down event (GPU returning to a lower-power state). Note the abrupt decline in GPU current and the near-sinusoidal PSU input current.

Where will this massive excess power go if there is a sudden drop in usage?

Problem 1:
Sub-second Load Swings

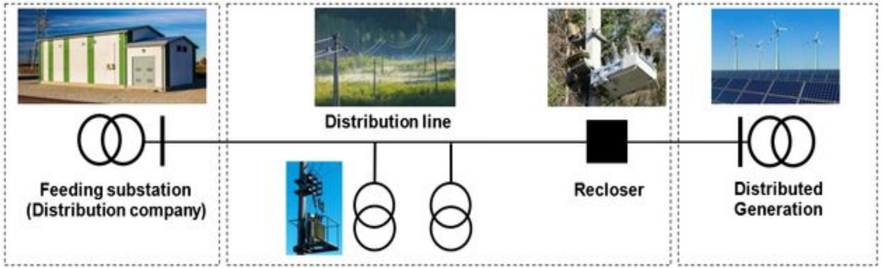


- **Unprecedented AI Load:** Shifting hundreds of megawatts in a fraction of a second breaks traditional grid management.
- **Slow Response Rates:** Traditional generators ramp too slowly (5-50 MW/min) to catch microsecond AI fluctuations.
- **Loss of Physical Inertia:** Historically, massive spinning turbines in power plants passively absorbed sudden load imbalances.
- **The Renewable Shift:** Wind and solar rely on inverters, which lack the physical mass needed to stabilize grid frequency.
- **New Infrastructure Required:** Active power quality devices (like STATCOMS) and batteries are now essential to compensate.

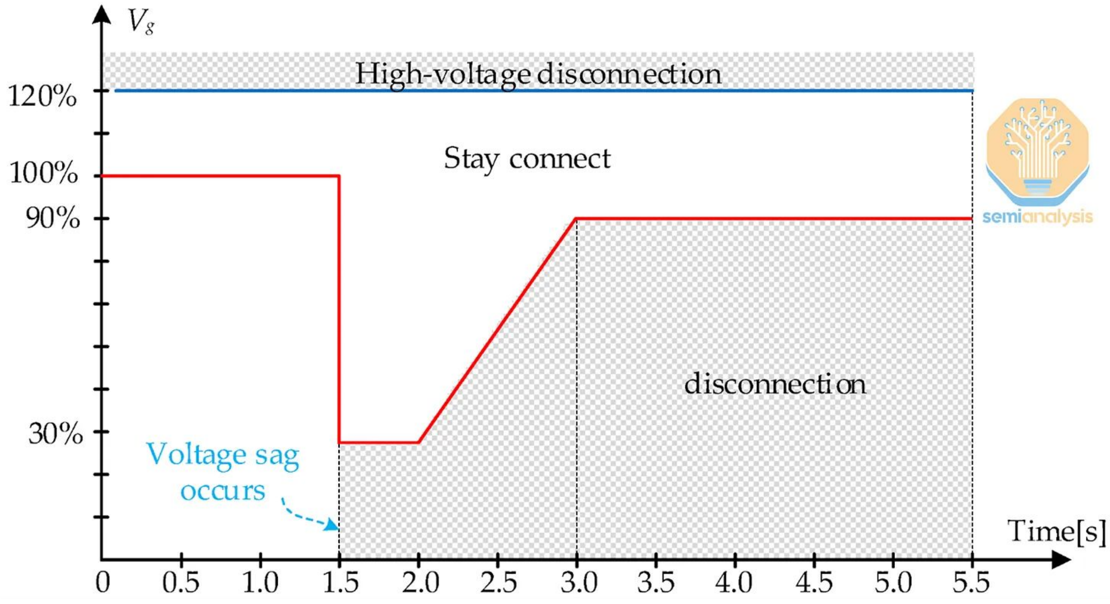
Problem 2: The LVRT Cascade Chain

LVRT is designed to ride through brief sags, but repeated disturbances can turn a local fault into a grid-wide event.

- **Recloser Cycles:** Grid breakers automatically trip and reset multiple times to clear distant physical line faults.
- **Voltage Sags:** These repeated breaker resets send temporary low-voltage shockwaves across the interconnected grid.
- **LVRT Explained:** "Low Voltage Ride-Through" is a datacenter's ability to survive these brief sags without disconnecting.
- **UPS Trigger:** If sags repeat, the datacenter's UPS permanently disconnects from the grid and switches to backup diesel.
- **Cascade Risk:** This sudden gigawatt load drop can crash other power plants and trigger a regional grid blackout.



Tavrida Electric



- **The Plunge:** Grid voltage suddenly crashes to 30% capacity.
- **The Safe Zone:** Datacenters *must* stay connected while above the red line.
- **The Disconnect:** Falling into the gray area forces a hard hardware shutdown

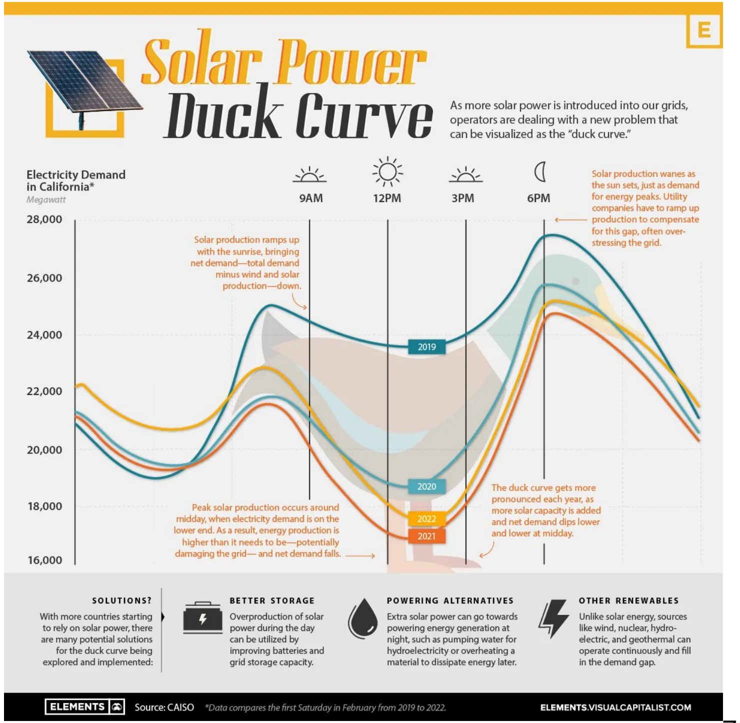
Scenario 1: ERCOT Stress Test — Datacenter Disconnection

Source: ERCOT May 2025 presentation by Yunzhi Cheng. Fault modeled: 345 kV transmission line in West Texas substation.

•Without SynCon: up to 2.5 GW at risk (HRML case). With SynCon: still 1.3–1.9 GW at risk — not enough

•Cost of SynCon: $30k–60k per MVA Reactive → ~$10M–20M per 1 GW datacenter

•West Texas DC load expected to soar past 10 GW rapidly



Source: Visual Capitalist

Does SynCon Help?

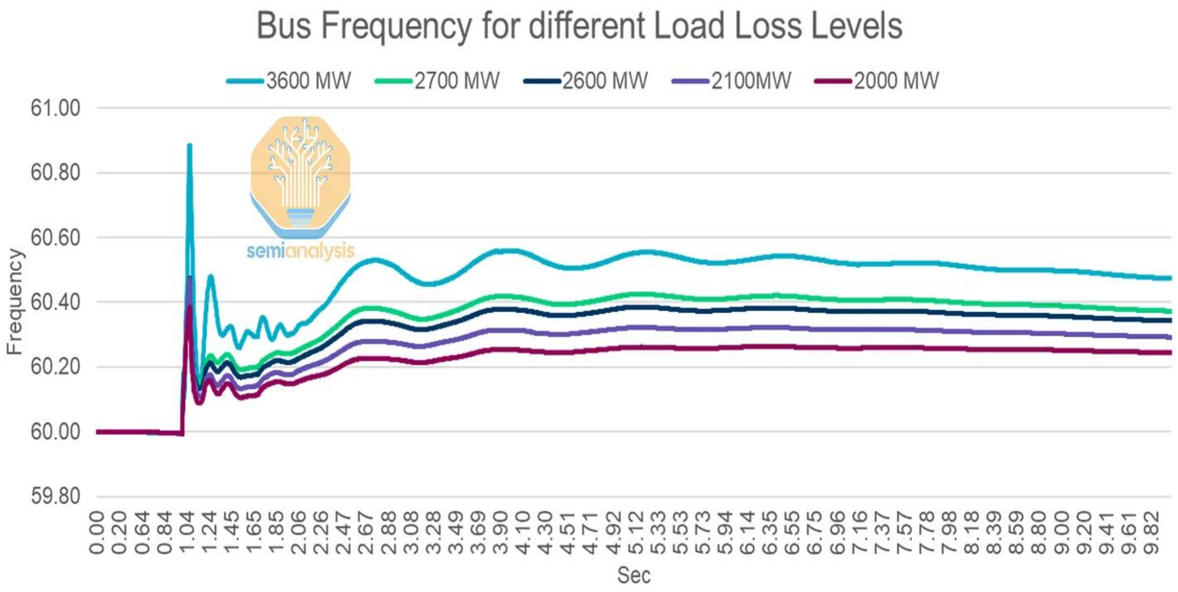
- Summer Peak: SynCon makes no difference — risk stays at ~1,500 MW either way
- High Renewable (HRML): SynCon helps — immediate trip risk drops from ~2,500 MW to ~1,500 MW
- But ~1,500 MW at risk still remains even with SynCon installed
- HRML = duck curve conditions = highest risk window, and SynCon isn't enough

Base Case Data Center Disconnection Risk		
Study Cases	Immediate Trip	LVRT 20 ms
Summer Peak (SP)	~1,500 MW	~1,500 MW
High Renewable Minimum Load (HRML)	~2,500 MW	~1,500 MW

Data Center Disconnection Risk + SynCon		
Study Cases	Immediate Trip	LVRT 20 ms
Summer Peak (SP)	~1,500 MW	~1,500 MW
High Renewable Minimum Load (HRML)	~2,500 MW	~1,500 MW

Source: ERCOT

Scenario 2: The Cascade risk



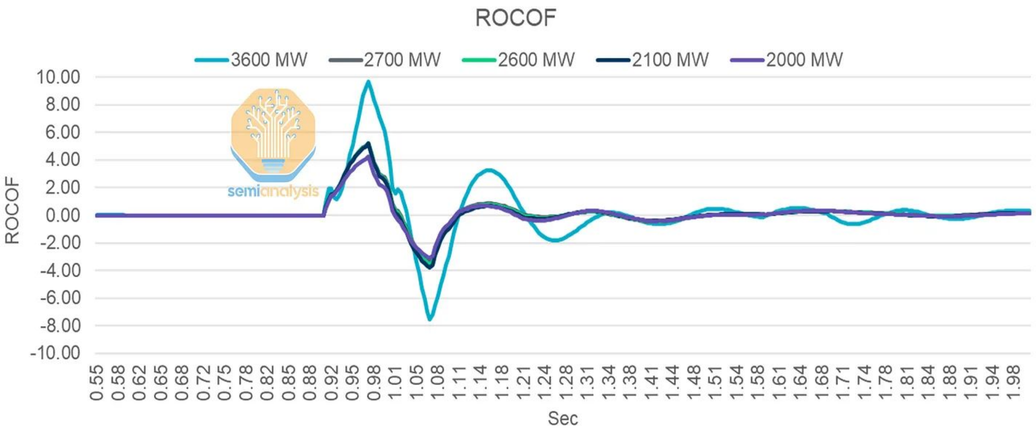
Source: ERCOT

- Each line = a different MW level of sudden datacenter load loss
- Below ~2,600 MW disconnection → frequency stays insecure
- ~1,950 MW is the safe threshold — anything above risks cascade

How Fast Does Frequency Fall?

- ROCOF = Rate of Change of Frequency — higher = more dangerous
- Above 2,600 MW loss → wide area voltage collapse across the grid"

MW Level	Max Absolute ROCOF
~3600	**9.70**
~2700	**5.23**
~2600	**5.21**
~2100	**5.18**
~2000	4.25



Scenario (Load Loss Level - MW)	Status
~3,600	Insecure - Wide Area Voltage Issues
~2,600	Insecure - Wide Area Voltage Issues
~2,450	Insecure - Wide Area Voltage Issues
~2,100	Insecure - Local Voltage Issues
~2,000	Insecure - Local Voltage Issues
~1,950	Secure

Source: ERCOT

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Scenario 3: The Iberian Warning -April 28, 2025

Real-world precedent from the report: a 27-second cascade collapse in Iberia.

1. **2.2 GW of generation trips offline**
   Miscalculated dispatch decisions by local grid operator
2. **Cascading voltage & frequency fluctuations**
   Breakers trip across Spain and Portugal
3. **Iberian grid isolated**
   Only weakly connected to the rest of Europe; external connections cannot stabilize
4. **Total collapse**
   Within 27 seconds

The Texas Parallel

Texas Interconnection has only 4 external connections. 2–2.5 GW of datacenter load tripping = same pathway. Could be triggered by a squirrel on the wrong power line near a West Texas substation.

How Do We Avoid the Nightmare?

The responsibility falls on the datacenter side. Hardware vendors have lined up with solutions — from supercapacitors to gigawatt-scale batteries. — SemiAnalysis

Solution: Tesla Megapack 2 XL & BESS

Tesla presented **Megapack 2 XL** at the ERCOT May 2025 meeting.

- **The BESS Solution:** Deploy gigawatt-scale batteries to buffer the grid.
- **Lightning Fast:** Charges and discharges hundreds of MWs in *seconds*.
- **All-in-One Architecture:** Fully self-contained with built-in inverters.
- **Thermal Management:** Integrated cooling survives extreme, repeated power cycling.
- **Massive Density:** Packs a staggering 200 MW of power into a single acre.

BESS (Battery Energy Storage System) — large-scale batteries co-located with the datacenter that absorb and release power in milliseconds, acting as a buffer between the AI cluster and the grid.

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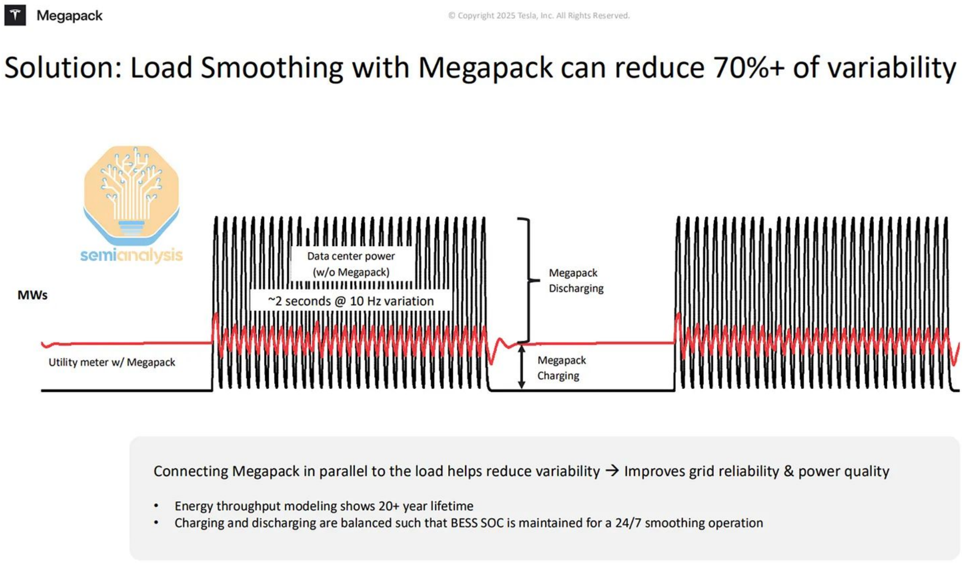
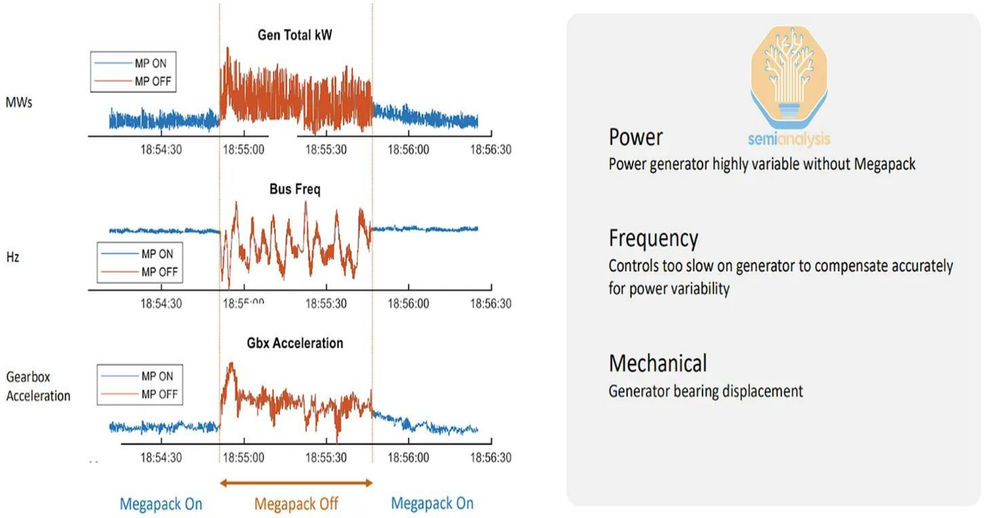
Load Smoothing: BESS in Action

- **Load Smoothing:** BESS acts as a massive buffer, absorbing or releasing power to flatten the volatile AI demand curve.
- **Massive Reduction:** This continuous charging and discharging eliminates over 70% of the datacenter's load variability.
- **The "Invisible" AI:** With BESS running (the red line), the utility company only ever sees a smooth, flat, "mimicked" load.
- **Outperforming Capacitors:** Capacitor banks are too limited; they cannot handle this load smoothing at the critical second-by-second scale.
- **Superior Flexibility:** BESS provides unmatched flexibility by managing power spikes across milliseconds, seconds, and minutes



Generator & Mechanical Protection (Megapack On vs. Off)

- **Fast Frequency Response:** BESS perfectly stabilizes electrical frequency, covering for backup generators that are too slow to react.
- **Protecting Local Generators:** Running BESS in tandem with onsite diesel generators massively smooths their total power output.
- **Mechanical Salvation:** By absorbing the chaotic power swings, BESS prevents violent mechanical wear (like bearing displacement) on turbine gearboxes.
- **Tandem Operation:** BESS isn't a replacement; it's designed to work seamlessly *with* existing UPS and generator infrastructure.
- **Solving LVRT:** If a voltage sag forces the datacenter’s UPS offline, the BESS instantly mimics the load to protect the grid from a cascading failure.



BESS vs Alternatives

	BESS	Capacitor Banks	Diesel Generators	Synchronous Condensers
Response time	ms–minutes	ms (but limited)	Minutes	Passive/instant
Load smoothing (seconds)	✓	✗	✗	Partial
LVRT support	✓	Partial	✗	✓
Demand response	✓	✗	✗	✗
Cost (1 GW DC)	~$1B	Lower	Lower	$10M–20M
Real deployment	xAI Colossus	Common	Common	ERCOT modeled

Is BESS the Best Solution?

Behind-the-meter BESS for datacenter load fluctuations.

The Case For

- Only technology that responds at ms, second, and minute timescales
- Already deployed: xAI Colossus + Tesla Megapack, Memphis
- Enables demand response without cutting load
- Improves generator lifespan through smoother operation
- BESS in series with generator = smoother ramp

The Caveats

- ~$1B for GW-scale: additional cost, not a replacement
- SOC must balance demand response vs. LVRT backup
- Utility-side demand response still immature
- ERCOT PCM (2026–27): incentives may be "a rounding error"
- Hardware alternatives like supercapacitors are explored in the paywalled section

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Key Takeaways

1 AI training load is a new class of grid threat

- Gigawatt-scale, millisecond swings — no historical precedent
- Meta's 30 MW cluster already flagged it; GW-scale is orders of magnitude worse

2 ERCOT has modeled the risk

- 2–2.5 GW disconnection → cascade blackout threshold
- Duck curve days are the highest-risk window

3 It has already happened — Iberian Peninsula, April 28, 2025

- 2.2 GW tripped → total collapse in 27 seconds
- Texas has only 4 external grid connections

4 BESS is the leading solution — but not a silver bullet

- Tesla Megapack deployed at xAI Colossus; responds at ms–minute timescale
- ~$1B for GW-scale; additional cost, not a replacement for UPS/diesel